CHALMERS

Finite Element Method - Structures, TME245

Computer assignment 2: Plate analysis

Instructions

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Preface

Within this computer exercise, roof panels for a garden shed, subjected to snow loading and temperature elevation, are to be analysed. The tasks will involve to implement a quadrilateral Kirchhoff plate element, to use that Kirchhoff plate element to analyse the deformations and stress distribution in the roof panels, and then finally to analyse the risk of buckling using second order Kirchhoff theory
The following MATLAB-files are made available to support this computer exercise:

- `rectMEsh.m`
  A function file that generates the mesh of the roof panels (plates), and provides the relevant matrices `Coord` and `Edof`. Please read additional instructions for how to use this in Python.

- `N_Kirchhoff_func.m`
  A function to compute shape functions for an isoparametric four noded plate element.

- `kirchhoff_funcs.m`
  A function to compute first and second shape function derivatives needed for the assignment.

Similarly, the following functions are available for Python users

- `N_Kirchhoff_func.py`
  A function to compute shape functions for an isoparametric four noded plate element.

- `kirchhoff_funcs.py`
  A function to compute first and second shape function derivatives needed for the assignment.

Learning outcomes: The learning outcomes of this project is to learn to implement a Kirchhoff plate element, to learn how to expand a standard Kirchhoff plate element to include second order in-plane effects (to allow for a linearise buckling analysis) and to conduct finite element simulations of plate problems, including post-processing of displacements and stresses.

Göteborg in February 2026

Martin Fagerström

Problem definition

A Chalmers professor wants to design and build his own shed for storing tools and garden equipment over the winter. As winter is coming, he is in a hurry and decides on a simple design where the roof is to be composed of a number of rectangular shaped aluminium plates mounted onto an underlying wooden support structure; see two examples of possible roof subdivisions in Figure 1 below (dashed lines indicate each roof plate). Each plate will be firmly attached to the support structure, such that it can be considered as clamped along all three edges (all degrees of freedom prescribed) and free to glide (see more details in the figure caption) along the fourth (being the one closest towards the ground).

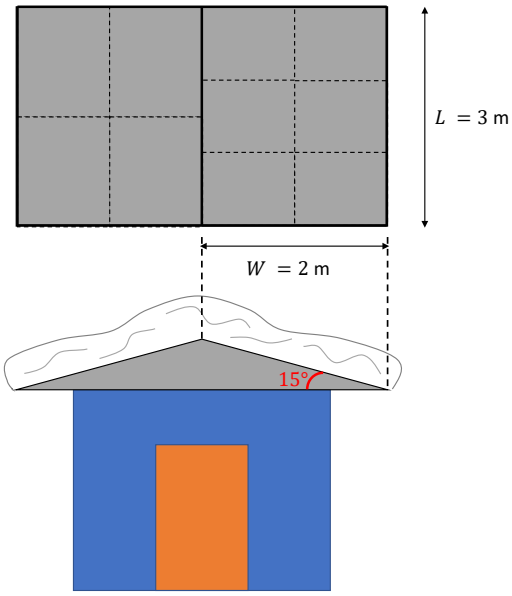

Figure 1: Simple sketch of garden shed analysed in Computer Assignment 2. Please note that possible subdivisions into smaller roof tapes are indicated with dashed lines.

The dimensions of the shed are such that the length of the roof is $L = 3\,\mathrm{m}$ and the projected half-width is $W = 2.0\,\mathrm{m}$, see Figure 1. Please note that the roof angle is 15° as also illustrated in the sketch. Furthermore, the roof plates is to be made from aluminium with the following properties: Young's modulus $E = 80\,\mathrm{GPa}$, Poisson's ratio $\nu = 0.2$, yield stress $\sigma_{\mathrm{yield}} = 200$ MPa and coefficient of thermal expansion $\alpha = 20 \times 10^{-6}\,1/\mathrm{K}$.

The design requirements are such that the safety factor against initial yielding or buckling (whichever is most critical) should be at least 2, and that the maximum deflection at any point is maximum 25 mm.

The loads to be considered are snow loads (Task 1, see also Figure 1) and elevated temperature (Task 2, see also Figure 3). Your job, is to determine suitable dimensions (width, length and thickness) of each roof plate such that the design criteria are all met.

Task 1: Linear elastic plate analysis using MATLAB or Python

Decide a suitable roof subdivision into smaller, *equally shaped* plates. For this task, please consider that the indicated subdivisions in Figure 1 most likely are too large to fulfill all requirements. At the same time, choosing to small plates introduces the need of a more dense underlying support structure. So please try to find a balance here.

For your choice of plate dimensions, consider a single roof plate as illustrated in Figure 2, subjected to a distributed pressure coming from *heavily compacted and wet* snow. The design criteria above must be fulfilled for a layer that is 0.25 m thick. Your job will now be to analyse this roof plate using Kirchhoff plate theory.

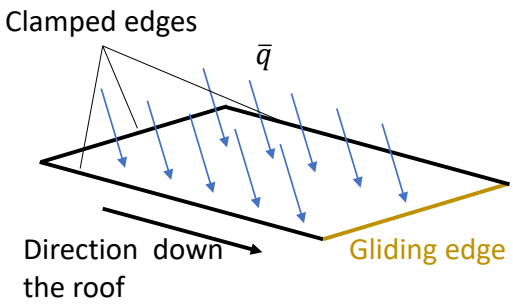

Figure 2: Single roof plate clamped along three edges (black) and free to glide (prescribed vertical displacements and rotation) along the fourth, subjected due to a distributed pressure load due to a layer of snow.

Complete the following tasks.

a) Define the loading and boundary conditions for a roof plate according to Kirchhoff plate theory.

b) Derive the expression for the element contributions to the external load vector(s) resulting from the snow load. Please consider the roof inclination of 15° as well as a heavily compacted and wet snow layer with a thickness of 0.25 m, and assume any other necessary parameters related to the load.

   As a starting point, the Voigt form of the total external work, including in-plane loads, is given by:

$$\begin{aligned}\delta W_{\text{ext}} &= \int_A \delta \underline{u}^{0^T} \underline{f}_u \, \mathrm{d}A + \int_{\mathcal{L}} \delta \underline{u}^{0^T} \underline{P} \, \mathrm{d}\mathcal{L} + \\ &\quad \int_A \delta w \, q \, \mathrm{d}A - \int_{\mathcal{L}} \frac{\partial(\delta w)}{\partial n} M_{nn} \, \mathrm{d}\mathcal{L} + \int_{\mathcal{L}} \delta w \left(V_n + \frac{\partial M_{nm}}{\partial m} \right) \mathrm{d}\mathcal{L}\end{aligned}$$

where $\delta \underline{u}^{0^T} = [\delta u_x^0 \quad \delta u_y^0]$ is the in-plane midplane displacement, $\underline{f}_u$ are distributed in-plane loads (or tractions) over A with dimension N/m^2 and $\underline{P}$ are distributed edge loads over $\mathcal{L}$ with unit N/m.

c) Describe the implementation procedure for a Kirchhoff plate element that considers both in-plane and out-of-plane loading and that outputs the total element stiffness matrix and the element external force contributions resulting from the snow load. Define any matrices you introduce.

d) Implement the element as a MATLAB or Python function file to be used to solve the problem. *Please check your implementation of the load vector contributions by comparing the total force components resulting from the applied load to the sum of computed nodal forces in each direction.*

e) Choose plate dimensions and write a small FE-program for plate analysis to solve the problem with the snow load.

To generate the mesh, please use the provided MATLAB-function `rectMesh` which generates the necessary matrices `Coord` and `Edof`, for which the degrees of freedom per node are ordered as $[u_x \; u_y \; w \; \theta_y \; \theta_x]$. If you implement this in Python, first run the function `rectMesh` in MATLAB and save the necessary data in a `.mat`-file of version 7. Saving the necessary data can be done via the command:

```
save('FileName.mat','-v7');
```

where `FileName.mat` is a file name of your choice. The data is then read in Python in the same way as for Computer Exercise 1.

Report if your plate dimensions satisfy the design criteria in terms of maximum vertical displacement. Also create plots that illustrate the in-plane and out-of-plane displacements. This can be done either by plotting the deformed mesh, or by making contour plots showing each displacement component separately.

f) Implement a MATLAB function that computes the non-zero stress components in the four in-plane locations of element integration points ($\xi_1 = \pm\frac{1}{\sqrt{3}}; \xi_2 \pm \frac{1}{\sqrt{3}}$ in the parent domain) for a given thickness coordinate.

Use this function to create three contour plots where, for each plot, each plate element is coloured by the in-plane average of the von Mises effective stress. The plots should be created for the three different thickness coordinates $z = -h/2$, $z = 0$ and $z = h/2$, where $h$is the thickness of the plate.

Please note that, in Voigt form, the von Mises stress σ_{vM} is found by:

$$\underline{s} = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{13} \\ \sigma_{23} \end{bmatrix} - \frac{(\sigma_{11} + \sigma_{22} + \sigma_{33})}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \sigma_{\text{vM}} = \sqrt{\frac{3}{2}\left(\underline{s}^T \underline{s} + \sigma_{12}^2 + \sigma_{13}^2 + \sigma_{23}^2\right)}.$$

Report if your plate dimensions also satisfy the stress conditions with the required safety factor.

g) If your plate dimensions are over-conservative or result in that one of the design criteria are not satisfied, then modify your design to a more appropriate one given the finite element code you have just developed. Please justify your final choice!

Task 2: Linearised buckling using MATLAB or Python

The next task is to compute the safety factor against buckling given a temperature elevation during a hot summer day, see Figure 3. For this case, please consider that the shed is built a bit hasty in October such that plates are mounted as stress free at a temperature of 10 °C. Furthermore, please consider that the roof plates can reach a maximum temperature of 40 °C during a hot summer day.

As such, the following tasks should be completed:

a) **Implement an element function (in MATLAB or Python) for a Kirchhoff plate bending element to be used for linearised pre-buckling analysis**. For this subtask, you may assume that the $\underline{\tilde{N}}^{\text{sec(R)}}$ is given as a constant input to the element.

Note that all code necessary for computing the $\underline{K}_{ww}^{K,e}$ matrix (see Lecture notes) is available in code snippets. Therefore, the task that remains is to implement the computation of $\underline{G}^{\text{e,(R)}}$ using a 3x3 Gauss integration scheme.

To clarify, the function file should have material properties, nodal coordinates, the thickness and a constant element $\underline{\tilde{N}}^{\text{sec(R)}}$ matrix as input, and have (at least) $\underline{K}_{ww}^{K,e}$ and $\underline{G}^{\text{e,(R)}}$ as output.

Describe the main steps of your implementation in the report.

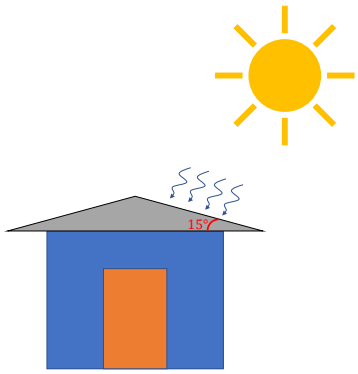

Figure 3: Simple sketch of garden shed subjected to sunlight.

b) Show that the element contribution to the *in-plane* external load vector ($\underline{f}_u^{\text{ext}}$), resulting from an elevated temperature ΔT, becomes:

$$\int_{A^e} \int_{-h/2}^{h/2} \alpha \Delta T \underline{B}^{eT} \underline{D} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \mathrm{d}A.$$

To do so, please use that the in-plane stresses can be computed as:

$$\underline{\sigma} = \underline{D} \left(\underline{\epsilon} - \underline{\epsilon}^{\text{th}} \right),$$

with the thermal strain $\underline{\epsilon}^{\text{th}} = \alpha \Delta T \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

Then implement a function that computes this contribution to the in-plane external load vector.

c) **Implement a small finite element programme for linearised pre-buckling analysis** that includes the following steps:

1. A solution of the in-plane deformation problem with the in-plane load coming from the elevated temperature $\Delta T = 30\,°\text{C}$. Here, you need to assemble the in-plane element stiffness $\underline{K}_{uu}^e$ into $\underline{K}_{uu}$, add the load contribution from the function implemented in subtask 2b into $\underline{f}_u^{ext}$ and then solve

$$\underline{K}_{uu}\, \underline{a}_u = \underline{f}_u^{ext}$$

with the proper in-plane displacement boundary conditions.

Note that all you need except for the thermal load contribution should have been implemented as part of solving Task 1.

2. For each element, evaluate the in-plane stress components (expand your code from Task 1f to also consider thermal strains as outlined above) in the element midpoint (in-plane midpoint and at $z = 0$) and compute
$$\underline{\tilde{N}}^{\mathrm{e,sec(R)}} \approx h \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix}$$
(h is the plate thickness) as an approximation of the in-plane loading that may trigger buckling. Then use this matrix as an element representative in-plane load[1] in the function file developed in Task 2a.
3. Use the function file created in Task 2a, with $\underline{\tilde{N}}^{\mathrm{e,sec(R)}}$ computed as described in the previous step, to compute $\underline{K}_{ww}^{K,e}$ and $\underline{G}^{\mathrm{e,(R)}}$ for each element. Assemble these contributions into $\underline{K}_{ww}^{K}$ and $\underline{G}^{(\mathrm{R})}$.
4. Find the smallest eigenvalue λ_1 and the corresponding buckling mode $\underline{z}_1$ from the generalised eigenvalue problem
$$\underline{K}_{wwFF}^{K}\underline{z}_i = -\lambda_i \underline{G}_{FF}^{(\mathrm{R})}\underline{z}_i$$
where $\underline{K}_{wwFF}^{K}$ and $\underline{G}_{FF}^{(\mathrm{R})}$ are the parts of $\underline{K}_{ww}^{K}$ and $\underline{G}^{(\mathrm{R})}$ associated with the free degrees of freedom.

Use this programme to **calculate the safety factor against buckling of the roof plate and plot the shape of the most critical buckling mode**. For the latter, plot the deformed shape using the corresponding out-of-plane degrees of freedom, this disregarding the degrees of freedom that define rotations.

[1]Please note that, in general, $\underline{\tilde{N}}^{\mathrm{sec(R)}}$ will vary inside the element. However, as this variation is often rather small we simplify the analysis by considering the midpoint resultants as representative constant values for the element.